

OilTrace AI

Technical Research Specification

Features • Datasets • Models • Literature Gaps • Technology Support • Evaluation

1. System Definition

OilTrace is an AI-assisted maritime oil-spill investigation and decision-support system combining satellite observations, image analysis, temporal reconstruction, environmental drift modelling, historical AIS reconstruction, evidence consistency and explainable evidence fusion. The project specification defines eight major features and does not treat the system as a legal attribution engine.

■filecite■turn3file3■L1-L20■

The SIH requirements include satellite-based spill detection/characterization, backward and forward slick modelling, historic AIS reconstruction/filtering/scoring and a visual investigation interface. ■filecite■turn4file3■L1-L20■

2. Literature-to-Implementation Framework

Existing-solution: establishes what has already been demonstrated.

Gap/limitation: establishes what remains difficult or incomplete.

Technology support: supports the technology selected to address the limitation.

OilTrace implementation: defines the integration and engineering choice.

Validation: measures OilTrace independently.

3. Dataset Architecture

SIH-provided satellite dataset

Primary project dataset for F1/F2. Exact file count, labels, resolution, polarization, geography and split must be read from the supplied raw archive.

Krestenitis Sentinel-1 benchmark

Supplementary public benchmark for detection/look-alike learning. Classes include sea, oil-spill, look-alike, ship and land.

SIH historical AIS (if supplied)

Primary vessel evidence for F4–F7. Expected fields: MMSI, timestamp, latitude, longitude, speed, course/heading and available metadata.

MarineCadastre AIS

Supplementary historical AIS benchmark for U.S. waters.

ERA5 hourly reanalysis

Historical environmental forcing for F3/F8, especially wind and related atmospheric variables.

Copernicus Marine currents

Ocean surface-current forcing for F3/F8.

Derived event/evidence dataset

Generated by F1–F4 and consumed by F5–F7: polygons, source probabilities, trajectories, residuals, evidence relations and provenance.

[ERA5 hourly dataset](#) provides hourly historical atmospheric variables; [Copernicus Marine current products](#) provide current fields; [MarineCadastre AIS tracks](#) provides historical AIS archives.

Dataset status: the uploaded project records identify the SIH satellite dataset and benchmark references, but the indexed File Library result did not expose the raw SIH dataset archive. Exact SIH image counts and file structure must therefore be taken directly from the supplied archive.

4. Feature Specifications

F1. AI Oil-Spill Detection & Look-Alike Analysis

Classification	Baseline + Improvement
Input data	Sentinel-1 SAR imagery; optional Sentinel-2 optical imagery; acquisition metadata and geolocation.
Dataset	Primary: SIH-provided satellite dataset after raw-file inventory. Supplementary: Krestenitis Sentinel-1 benchmark containing sea, oil-spill, look-alike, ship and land classes.
Model / technology	U-Net baseline. Candidate alternatives: U-Net++, DeepLabV3+, SegFormer. Optional attention/look-alike classifier only if justified by validation.
Input features / variables	SAR backscatter (VV/VH where available), texture, local context, optional optical bands, acquisition geometry, temporal context.
Output	Oil probability; segmentation mask; georeferenced polygon; confidence; look-alike risk; data-quality flag.
Evaluation	IoU, Dice/F1, precision, recall, false-positive and false-negative rate; scene/region-held-out testing.

Literature mapping

Existing — [A Deep-Learning Framework for the Detection of Oil Spills from SAR Data](#)

Deep-learning SAR detection/segmentation using a two-stage framework.

Gap — [Marine oil spill detection and segmentation in SAR data with two steps Deep Learning framework](#)

Look-alikes and false positives remain relevant to automated SAR detection.

Technology support — [Oil Spill Identification based on Dual Attention UNet Model Using Synthetic Aperture Radar Images](#)

Supports attention-enhanced U-Net for SAR.

Research gap

Existing DL methods establish feasibility. The remaining problem is reliable detection across scenes and explicit handling of look-alike uncertainty.

OilTrace implementation

Train a baseline first; introduce architectural improvements only when ablation demonstrates measurable benefit. Output a georeferenced evidence object for downstream stages.

F2. Multi-Temporal Spill Reconstruction & Characterization

Classification	Improvement
Input data	Multiple timestamped Sentinel-1 observations covering the same or overlapping area of interest.
Dataset	Primary: SIH event observations. Supplementary public oil-spill scenes for sequence/geometry validation where compatible.
Model / technology	Segmentation output from F1 + GIS geometry extraction. Temporal feature tracking; optional lightweight temporal model only if sequence labels are sufficient.
Input features / variables	Area, perimeter, centroid, bounding box, major/minor axes, orientation, shape descriptors, polygon IoU, centroid displacement, area change, persistence, observation gap.
Output	Timestamped observed spill states; geometry table; progression timeline; derived change indicators.
Evaluation	Polygon IoU; centroid and area error; temporal continuity; track continuity; comparison with known event sequences.

Literature mapping

Existing — [Marine oil spill detection and segmentation in SAR data with two steps Deep Learning framework](#)

Provides spill segmentation and boundary information.

Gap — [Monitoring Oil Spill Progression and Oil Spill Volume Using Satellite Images](#)

Demonstrates quantitative progression and characterization from satellite imagery/GIS.

Technology support — [Oil spills detection from SAR Earth observations based on a hybrid CNN transformer networks](#)

Supports richer SAR representation for segmentation and boundaries.

Research gap

A single image represents one observation. Source investigation requires an explicit temporal sequence and a distinction between observed and predicted states.

OilTrace implementation

Store every satellite observation as a timestamped state and derive geometry/change features before drift and AIS analysis.

Predicted/interpolated states are never labelled as observations.

F3. Environmental Drift & Backward Hindcasting → Source Hypotheses

Classification	Existing + Improvement
Input data	Observed spill polygons/timestamps; historical surface currents; wind fields; optional waves; particle/model parameters.
Dataset	SIH satellite event data + timestamp-matched environmental forcing. Recommended historical sources: ERA5 winds and Copernicus Marine surface currents.
Model / technology	Lagrangian particle tracking / OpenDrift-style model. Backward ensemble or probabilistic hindcasting.
Input features / variables	Particle latitude/longitude/time, current and wind velocity, diffusion, ensemble member, backtracked distance, source probability density, uncertainty radius.
Output	Candidate source regions; plausible origin-time window; particle trajectories; uncertainty envelope.
Evaluation	Source-region coverage/hit rate where ground truth exists; trajectory error; sensitivity to forcing resolution and ensemble spread.

Literature mapping

Existing — [A surface Lagrangian algorithm applied to the 2019 South Atlantic oil spill](#)

Uses backward/forward Lagrangian simulations for source investigation.

Gap — [Performance of a simple backtracking method for marine oil source searching in a 3D ocean](#)

Shows sensitivity to ocean dynamics, backtracking time and particle assumptions.

Technology support — [Bayesian identification of oil spill source parameters from image contours](#)

Supports probabilistic source estimation.

Research gap

Backtracking is sensitive to environmental forcing, observation timing, diffusion and oil properties; a single point estimate can overstate certainty.

OilTrace implementation

Generate multiple source hypotheses and an uncertainty envelope. The output is a hypothesis for AIS filtering, not proof of source.

F4. Historical AIS Vessel Reconstruction & Correlation

Classification	Baseline / Existing
Input data	Historical AIS: MMSI, timestamp, latitude, longitude, speed, course/heading and available vessel metadata.
Dataset	Primary: SIH historical AIS if supplied. If additional benchmark data are required, use a geographically appropriate historical AIS source such as MarineCadastre for U.S. waters.
Model / technology	Geospatial and temporal filtering; trajectory reconstruction; compatibility features. Optional anomaly model only if adequate data exist.
Input features / variables	Distance to source region, temporal compatibility, track overlap, speed/course compatibility, track completeness, vessel type, gaps.
Output	Reconstructed vessel tracks; filtered candidate vessels; compatibility feature table.
Evaluation	Candidate recall; filtering reduction; track completeness; reconstruction error; temporal/spatial compatibility.

Literature mapping

Existing — [A new ship tracing technology from oil spills based on multi-source data](#)

Combines SAR, AIS and environmental/drift information for vessel tracing.

Gap — [Oil pollution in the Eastern Arabian Sea from invisible sources: A multi-technique approach](#)

Demonstrates the need for multiple evidence types in source interpretation.

Technology support — [Satellite imagery in evaluating oil spill modelling scenarios for the Syrian oil spill crisis, summer 2021](#)

Supports integration of satellite observations with modelled spill scenarios.

Research gap

AIS-based tracing exists. The remaining system problem is transparent connection of historical vessel tracks to uncertain source windows and incomplete/conflicting observations.

OilTrace implementation

Use source-hypothesis windows to filter AIS and calculate interpretable compatibility features before evidence ranking.

F5. Cross-Source Consistency & Evidence Conflict Detection

Classification	Proposed Differentiator
Input data	Spill observations, temporal states, environmental compatibility, source hypotheses, AIS candidate evidence and data-quality metadata.
Dataset	Derived evidence dataset generated from F1–F4; each record retains source, timestamp, geometry, confidence and provenance.
Model / technology	Rule/constraint engine for MVP. Optional learned conflict classifier only if labelled conflict examples are available.
Input features / variables	Support/contradict/unknown state; spatial residual; temporal residual; drift residual; AIS gap ratio; sensor confidence; observation count; forcing quality.
Output	Evidence matrix/graph with SUPPORTS, CONTRADICTS and UNKNOWN relations.
Evaluation	Expert-reviewed consistency precision/recall; false escalation rate; calibration of evidence states.

Literature mapping

Existing — [A new ship tracing technology from oil spills based on multi-source data](#)

Direct precedent for combining SAR, AIS and environmental information.

Gap — [Assessing ocean ensemble drift predictions by comparison with observed oil slicks](#)

Demonstrates meaningful mismatch between model predictions and observations.

Technology support — [Bayesian identification of oil spill source parameters from image contours](#)

Supports explicit probabilistic treatment of uncertain evidence.

Research gap

Multi-source approaches can collapse heterogeneous evidence into a final result without preserving which sources disagree or which information is missing.

OilTrace implementation

Represent support, contradiction and unknown states explicitly and preserve provenance for every evidence item.

F6. Evidence Fusion & Dynamic Hypothesis Ranking

Classification	Proposed System-Level Differentiator
Input data	Evidence matrix from F5; candidate-vessel features; source-hypothesis probability; sensor/model/data-quality factors.
Dataset	Derived evidence dataset. Supervised attribution labels are required only if a learned ranking model is introduced.
Model / technology	Transparent weighted score for MVP; Bayesian updating/calibrated probabilistic ranking as an advanced option after validation.
Input features / variables	Spatial/time/drift compatibility, AIS completeness, behavioural indicators, sensor confidence, contradiction count, data-quality weight.
Output	Ranked candidate hypotheses; component-wise explanation; confidence band; insufficient-evidence outcome.
Evaluation	Recall@K, MRR/NDCG where ground truth exists; calibration; sensitivity analysis; expert assessment.

Literature mapping

Existing — [A new ship tracing technology from oil spills based on multi-source data](#)

Uses trajectory similarity to identify likely vessels.

Gap — [Bayesian identification of oil spill source parameters from image contours](#)

Shows value of probabilistic source inference.

Technology support — [Oil pollution in the Eastern Arabian Sea from invisible sources: A multi-technique approach](#)

Supports multi-technique evidence interpretation.

Research gap

Composite scoring can hide why a candidate ranks highly and can overstate certainty when inputs are incomplete or correlated.

OilTrace implementation

Keep component evidence visible, weight data quality, update rankings with new observations, and allow an unresolved/insufficient-evidence result.

F7. Forensic Investigation Graph & Explainable Evidence Chain

Classification	Proposed Workflow Differentiator
Input data	All observations, hypotheses, environmental states, AIS candidates, scores, evidence relations and provenance metadata.
Dataset	Derived event graph; each node/edge references an input observation or computed result.
Model / technology	Graph/provenance model. PostgreSQL/PostGIS can implement the MVP without requiring a separate graph database.
Input features / variables	Node types: spill observation, source hypothesis, environmental state, vessel, evidence item, forecast. Edge types: DERIVED-FROM, SUPPORTS, CONTRADICTS, TEMPORALLY-COMPATIBLE.
Output	Clickable evidence chain; candidate explanation; timeline/map trace; audit trail.
Evaluation	Traceability completeness; explanation correctness; investigator task completion/time-to-understand.

Literature mapping

Existing — [A new ship tracing technology from oil spills based on multi-source data](#)

Provides a multi-source tracing foundation.

Gap — [SIH26143 problem statement](#)

Requires a suitable visual interface for investigation and candidate assessment.

Technology support — [Satellite imagery in evaluating oil spill modelling scenarios for the Syrian oil spill crisis, summer 2021](#)

Supports interpretation of satellite observations alongside model scenarios.

Research gap

Research outputs commonly end at maps, model metrics or rankings. Investigation requires a traceable chain linking evidence to each hypothesis.

OilTrace implementation

Represent the investigation as an auditable evidence graph with provenance and explicit relation types.

F8. Forward Forecasting, Impact Assessment & Historical Replay

Classification	Existing + Improvement + Validation
Input data	Latest confirmed spill state; wind/current/wave forcing; forecast horizon; particle parameters; geographic impact layers.
Dataset	Historical satellite events + timestamp-matched environmental forcing for replay/validation. Operational forcing may be added later.
Model / technology	OpenDrift/OpenOil-style forward Lagrangian ensemble; GIS impact overlays.
Input features / variables	Particle position/time, forcing variables, ensemble spread, distance to coastline/sensitive zones, forecast confidence.
Output	Forecast trajectory envelope; impact-area candidates; historical replay comparison.
Evaluation	Trajectory error; observed-region coverage; ensemble calibration; forecast skill versus horizon.

Literature mapping

Existing — [An ocean-wave-trajectory forecasting system for the eastern Baltic Sea](#)

OpenDrift forecasting with environmental forcing and validation.

Gap — [Assessing ocean ensemble drift predictions by comparison with observed oil slicks](#)

Shows uncertainty and forcing differences affect forecast agreement.

Technology support — [Operational Modeling of North Aegean Oil Spills Forced by Real-Time Met-Ocean Forecasts](#)

Supports operational forward modelling with real-time met-ocean forcing.

Research gap

Forecast accuracy depends on forcing quality, initial state, oil properties and forecast horizon; exact future paths cannot be guaranteed.

OilTrace implementation

Present forecast envelopes/scenarios and validate historical forecasts against subsequent satellite observations.

5. Model and Technology Stack

Detection / segmentation	U-Net baseline; compare U-Net++ / DeepLabV3+ / SegFormer only where validation justifies.
Look-alike handling	Multi-class segmentation or secondary classifier; optional attention module after ablation.
Temporal reconstruction	GIS geometry + temporal feature tracking; optional sequence model only if sufficient labelled sequences exist.
Backward source modelling	Lagrangian / OpenDrift-style ensemble hindcasting.
AIS reconstruction	Spatial-temporal filtering + trajectory compatibility features.
Consistency layer	Explicit rules for SUPPORTS / CONTRADICTS / UNKNOWN.
Evidence ranking	Transparent weighted score; probabilistic update only after validation.
Evidence graph	PostgreSQL/PostGIS provenance model for MVP.
Forward forecast	OpenDrift/OpenOil-style forward ensemble.

6. Training Data vs Derived Data

Training data: labelled satellite images and masks used for vision models.

Environmental forcing: wind/current fields used by the physical drift model.

AIS evidence: historical vessel observations used primarily for filtering and correlation.

Derived data: polygons, temporal states, source probability maps, drift trajectories, candidate features and evidence relations.

7. Data Preparation and Validation

Satellite: preprocessing → normalization → tiling → label alignment → scene/event/region split → training-only augmentation.

Environmental: timestamp matching → spatial interpolation → unit/provenance checks.

AIS: validation → time sorting → track construction → gap identification → source-window filtering → compatibility features.

Evidence: preserve source, timestamp, model/version, confidence and uncertainty.

Validation controls

Use scene/event/region-level splits where possible; avoid distributing adjacent tiles from one scene across train and test.

Do not use future observations to construct earlier-time features.

Do not treat interpolated AIS positions as observed transmissions.

Match historical environmental forcing to the historical timestamp during replay/hindcasting.

Do not train a ranking model on labels generated by the same ranking system.

Report published-paper performance as literature context only; report OilTrace performance separately.

8. Research Contribution

Established components include satellite oil-spill detection, segmentation, Lagrangian drift modelling, AIS-based vessel tracing, GIS analysis and operational forecasting. The proposed system-level contribution is their integration into an auditable investigation workflow that maintains multiple source hypotheses, preserves supporting/contradictory/missing evidence, incorporates data quality and uncertainty into ranking, and provides a traceable evidence chain.

9. Complete Reference Library

[A Deep-Learning Framework for the Detection of Oil Spills from SAR Data](#)

Deep-learning SAR detection/segmentation using a two-stage framework.

[Marine oil spill detection and segmentation in SAR data with two steps Deep Learning framework](#)

Look-alikes and false positives remain relevant to automated SAR detection.

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10. Implementation Checklist

Dataset

Inventory the raw SIH dataset: filenames, formats, image count, mask/label count, classes, resolution, polarization, geography, timestamps, split and licence.

Models

Establish the U-Net baseline; test look-alike handling; implement GIS characterization; connect environmental forcing; implement backward Lagrangian ensemble; implement historical AIS filtering; implement evidence consistency/ranking; implement evidence provenance; add forecasting/replay where validated.

11. Technical Boundaries

OilTrace should report probable/possible source hypotheses and candidate-vessel associations, not legal responsibility. Backward modelling is uncertain; AIS records are historical positioning evidence; forecasts are scenario-based; and model performance depends on sensor coverage, data quality, geographic domain and environmental forcing.